

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Constraint Satisfaction and Propagation

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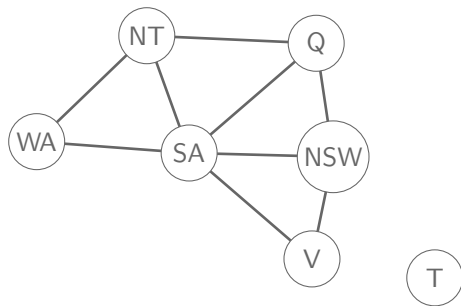
University of Arkansas

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Today's Topics

1. **CSP Structure and Formulation:** factored states, variables, domains, constraints, and representative problems
2. **Constraint Propagation:** node consistency, arc consistency, REVISE, and AC-3
3. **Strength and Limits:** cascading domain reduction, local versus global reasoning, and inference cost
4. **Backtracking, Inference, and CSP Heuristics:** ordering choices, inference during search, and learning from failure

The Map-Coloring Problem

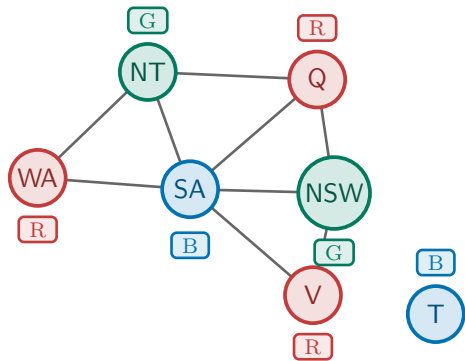


The task

1. Each circle represents an Australian region.
2. Each line represents a shared land border.
3. Give every region one color: **R**, **G**, or **B**.

Regions joined by a line must have different colors.

The Map-Coloring Problem



A valid coloring

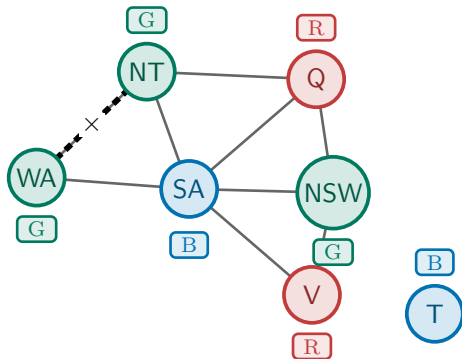
$$\begin{aligned} WA &= \text{R}, & NT &= \text{G}, & SA &= \text{B}, \\ Q &= \text{R}, & NSW &= \text{G}, & V &= \text{R}, & T &= \text{B}. \end{aligned}$$

Every pair connected by a line has different colors.

All border rules are satisfied.

Tasmania has no graph edges, so any of the three colors is legal for T .

The Map-Coloring Problem



An invalid coloring

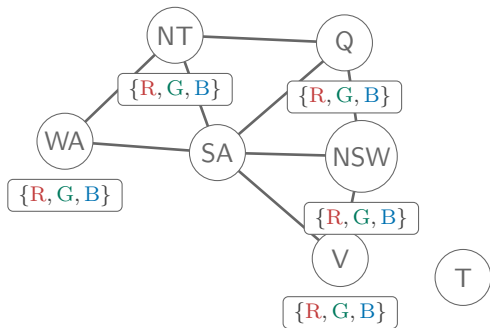
Change only WA from R to G .

Now $WA = NT = G$ even though the two regions share a border.

Every region is colored, but one border rule is broken.

One same-color edge makes the coloring invalid.

One Assignment Shrinks Five Domains



Opening question

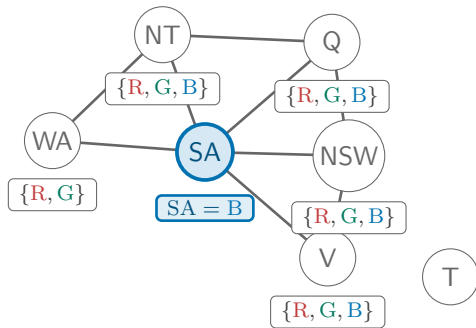
Each of SA's five neighbors initially has three color choices.

After assigning $SA = B$, how many combinations remain when we apply only SA's five incident edges?

243 96 32 1

Each neighbor starts with the choices $\{R, G, B\}$.

One Assignment Shrinks Five Domains



Opening question

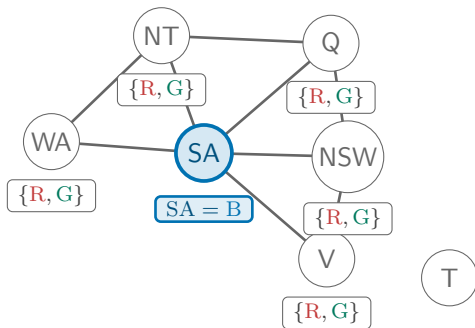
Each of SA's five neighbors initially has three color choices.

After assigning $SA = B$, how many combinations remain when we apply only SA's five incident edges?

243 96 32 1

The edge $WA \neq SA$ removes B from WA's choices.

One Assignment Shrinks Five Domains



Opening question

Each of SA's five neighbors initially has three color choices.

After assigning SA = **B**, how many combinations remain when we apply only SA's five incident edges?

243 96 **32** 1

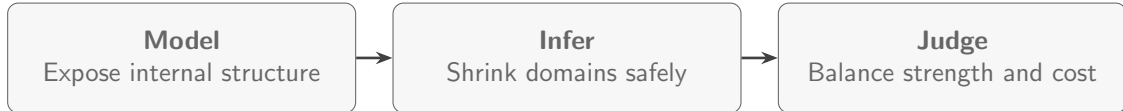
$$2 \times 2 \times 2 \times 2 \times 2 = \mathbf{32}$$

These 32 precede checking edges among the five neighbors themselves.

Learning Objectives

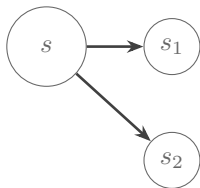
By the end of this deck, you should be able to:

1. **Formulate** a finite-domain problem using variables, domains, and constraints.
2. **Distinguish** partial, complete, and consistent assignments and identify a solution.
3. **Trace** node consistency, arc consistency, and the main loop of AC-3.
4. **Explain** why stronger propagation can prune more yet still fail to prove global satisfiability.



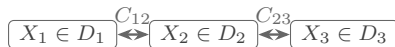
From Atomic to Factored States

Atomic state-space search



The algorithm sees an indivisible state and a domain-specific successor function.

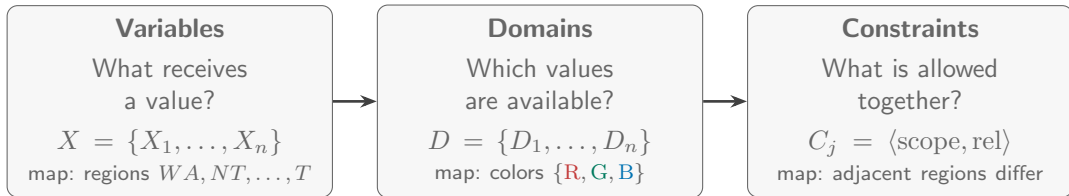
Factored CSP state



The algorithm can inspect variables, legal values, and the constraints connecting them.

Structure supports **general-purpose** pruning rules.

CSP Components



Model the choices first; propagation exploits their structure.

CSP Structure in Different Problems

Queens

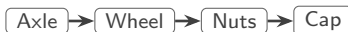
	Q		
			Q
Q			
		Q	

X: Q_i = row chosen in column i .

D: $D_i = \{1, \dots, n\}$.

C: $Q_i \neq Q_j$ and $|Q_i - Q_j| \neq |i - j|$.

Job-shop scheduling



X: operation start times.

D: feasible time slots.

C: precedence, duration, resources, and deadlines.

Sudoku

		3		2		6		
9			3		5			1
		1	8		6	4		
		8	1		2	9		
7								8
		6	7		8	2		
		2	6		9	5		
8			2		3			9
		5		1		3		

X: one value per cell.

D: $\{1, \dots, 9\}$, restricted by clues.

C: Alldiff for every row, column, and box.

Every application becomes the same template: choose variables X , domains D , and constraints C .

Cryptarithmic Formulation

$$\begin{array}{r} \\ \\ + \\ \hline F \end{array}$$

Hard example

What must a generic solver know besides *try digits*?

Variables and domains

$$\{F, T, U, W, R, O, C_1, C_2, C_3\}$$

$$F, T, U, W, R, O \in \{0, \dots, 9\}$$

$$C_1, C_2, C_3 \in \{0, 1\}$$

$$F \neq 0, \quad T \neq 0$$

$$\text{Alldiff}(F, T, U, W, R, O)$$

Column constraints

$$O + O = R + 10C_1,$$

$$C_1 + W + W = U + 10C_2,$$

$$C_2 + T + T = O + 10C_3,$$

$$C_3 = F.$$

Carries convert one global equation into local structure.

Assignments and Solutions

Two independent questions classify an assignment: how much has been assigned, and do the choices conflict?

Coverage of the Assignment

Partial assignment Only some variables have values.

$$\{WA = \text{R}, NT = \text{G}\}$$

Complete assignment Every variable has a value—for example, all seven regions are colored.

Solution

A **solution** is an assignment that is both **complete** and **consistent**.
Search extends partial assignments toward this goal.

Constraint Satisfaction So Far

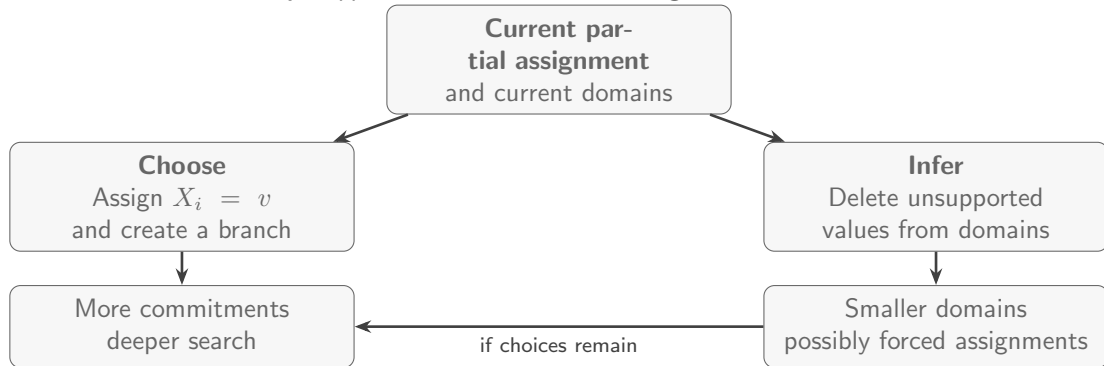
Consistent assignment The choices made so far violate no constraint.

$$\{WA = \text{R}, NT = \text{G}\} \text{ satisfies } WA \neq NT.$$

Contrast $\{WA = \text{R}, NT = \text{R}\}$ is not consistent because it violates $WA \neq NT$.

Inference within CSP Search

Roadmap: support test $\rightarrow$ **REVISE** $\rightarrow$ neighbor recheck $\rightarrow$ AC-3



Preprocessing

Propagate once before search begins.

Interleaving

Propagate again after each assignment.

Safe Deletion: Node and Arc Consistency

Node consistency

Every value satisfies the variable's unary constraints.

$$D_{SA} = \{\text{R}, \text{G}, \text{B}\}, \quad SA \neq \text{G}$$

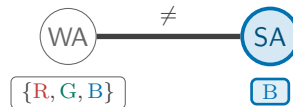
$$D_{SA} \leftarrow \{\text{R}, \text{B}\}$$

One node; inspect unary constraints.

Arc consistency

X_i is arc-consistent with X_j when

$$\forall x \in D_i, \exists y \in D_j : C_{ij}(x, y).$$



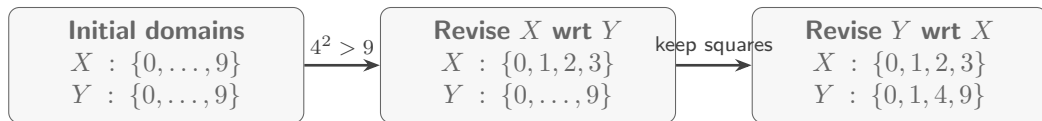
$$D_{WA} : \{\text{R}, \text{G}, \text{B}\} \longrightarrow \{\text{R}, \text{G}\}$$

$SA = \text{B}$ leaves WA 's blue value without support.

REVISE(X_i, X_j): delete from D_i every value without support in D_j .

REVISE Trace: $Y = X^2$

$$D_X = D_Y = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}, \quad C(X, Y) : Y = X^2.$$



Deleted from X

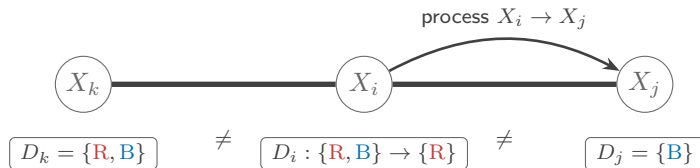
4, ..., 9 have no square in D_Y .

Deleted from Y

2, 3, 5, 6, 7, 8 are not squares of a remaining X .

A domain change can invalidate support on adjacent arcs.

Neighbor Rechecks after Domain Revision



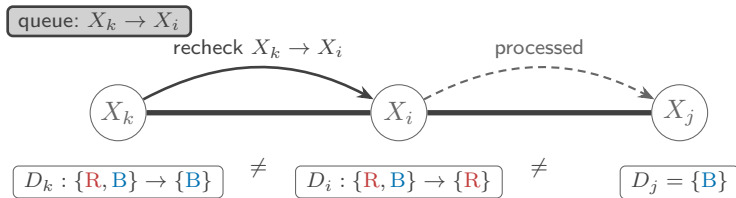
1. Revise $X_i \rightarrow X_j$

$\text{B} \in D_i$ has no support in $D_j = \{\text{B}\}$, so delete it:

$$D_i : \{\text{R}, \text{B}\} \longrightarrow \{\text{R}\}.$$

Queue rule: if D_i changes, re-enqueue $X_k \rightarrow X_i$ from every other neighbor X_k .

Neighbor Rechecks after Domain Revision



1. Revise $X_i \rightarrow X_j$

$\mathbf{B} \in D_i$ has no support in $D_j = \{\mathbf{B}\}$, so delete it:

$$D_i : \{\mathbf{R}, \mathbf{B}\} \longrightarrow \{\mathbf{R}\}.$$

2. Recheck $X_k \rightarrow X_i$

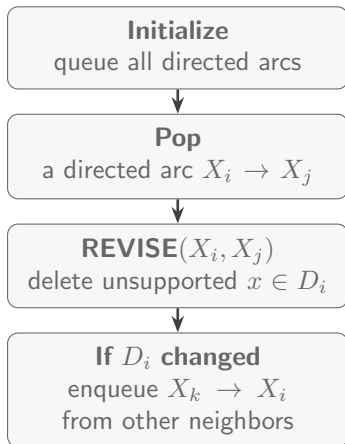
After D_i shrinks, $\mathbf{R} \in D_k$ loses its support:

$$D_k : \{\mathbf{R}, \mathbf{B}\} \longrightarrow \{\mathbf{B}\}.$$

Queue rule: if D_i changes, re-enqueue $X_k \rightarrow X_i$ from every other neighbor X_k .

AC-3: Queue-Based Arc Revision

Purpose: repeat **REVISE** on queued arcs until the queue is empty or a domain becomes empty.



Repeat until the queue is empty or a domain is empty.

Stop conditions

- Some $D_i = \emptyset$: failure.
- Queue empty: arc-consistent; search may still be needed.

What is preserved?

Only unsupported values are removed, so every solution is preserved.

Worst-case time

For c arcs and maximum domain size d :

$$O(cd^3).$$

Sudoku Propagation Cascade

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					?			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

Try it first

Which value survives for $E6$ after its box and column are checked?
Then follow column 6 to infer $I6$ and $A6$.

Cascade

The three cells become singletons in sequence:

$$E6 = ? \implies I6 = ? \implies A6 = ?.$$

Sudoku Propagation Cascade

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
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F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	?	3		

1. **Cell $E6$.** Its box contains 1, 2, 7, 8; column 6 contains 2, 3, 5, 6, 8, 9.

$$D_{E6} = \{1, \dots, 9\} \setminus \{1, 2, 3, 5, 6, 7, 8, 9\} = \boxed{\{4\}}.$$

Cascade

The three cells become singletons in sequence:

$$E6 = 4 \implies I6 = ? \implies A6 = ?.$$

Sudoku Propagation Cascade

	1	2	3	4	5	6	7	8	9
A			3		2	?	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
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F			6	7		8	2		
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$$D_{E6} = \{1, \dots, 9\} \setminus \{1, 2, 3, 5, 6, 7, 8, 9\} = \boxed{\{4\}}.$$

2. **Cell $I6$.** With $E6 = 4$, column 6 rules out 2, 3, 4, 5, 6, 8, 9; $I5 = 1$ rules out 1.

$$D_{I6} = \{1, \dots, 9\} \setminus \{1, 2, 3, 4, 5, 6, 8, 9\} = \boxed{\{7\}}.$$

Cascade

The three cells become singletons in sequence:

$$E6 = 4 \implies I6 = 7 \implies A6 = ?.$$

Sudoku Propagation Cascade

	1	2	3	4	5	6	7	8	9
A			3		2	1	6		
B	9			3		5			1
C			1	8		6	4		
D			8	1		2	9		
E	7					4			8
F			6	7		8	2		
G			2	6		9	5		
H	8			2		3			9
I			5		1	7	3		

1. **Cell E6.** Its box contains 1, 2, 7, 8; column 6 contains 2, 3, 5, 6, 8, 9.

$$D_{E6} = \{1, \dots, 9\} \setminus \{1, 2, 3, 5, 6, 7, 8, 9\} = \boxed{\{4\}}.$$

2. **Cell I6.** With $E6 = 4$, column 6 rules out 2, 3, 4, 5, 6, 8, 9; $I5 = 1$ rules out 1.

$$D_{I6} = \{1, \dots, 9\} \setminus \{1, 2, 3, 4, 5, 6, 8, 9\} = \boxed{\{7\}}.$$

3. **Cell A6.** After $I6 = 7$, column 6 contains 2, 3, 4, 5, 6, 7, 8, 9.

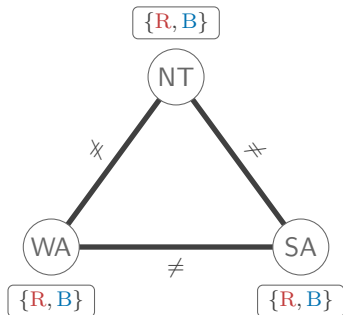
$$D_{A6} = \{1, \dots, 9\} \setminus \{2, 3, 4, 5, 6, 7, 8, 9\} = \boxed{\{1\}}.$$

Cascade

The three cells become singletons in sequence:

$$E6 = 4 \implies I6 = 7 \implies A6 = 1.$$

Limits of Arc Consistency



Every arc has support

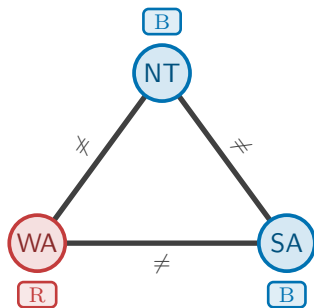
Across any edge:

$\textcolor{red}{R}$ is supported by $\textcolor{blue}{B}$, $\textcolor{blue}{B}$ is supported by $\textcolor{red}{R}$.

AC-3 deletes nothing.

Does that prove a solution exists?

Limits of Arc Consistency



One choice forces two neighbors

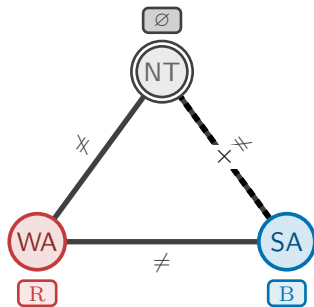
Choose $WA = R$.

$$D_{NT} : \{R, B\} \rightarrow \{B\}, \quad D_{SA} : \{R, B\} \rightarrow \{B\}.$$

Both neighbors must be blue because each differs from WA .

What happens on the remaining edge
 $NT \neq SA$?

Limits of Arc Consistency



The remaining arc exposes failure

NT 's only value B has no different-color support in $D_{SA} = \{B\}$.

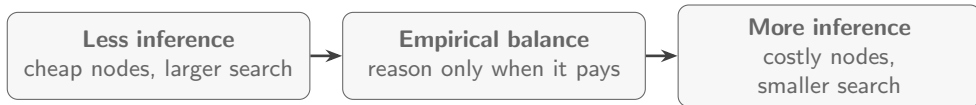
$$D_{NT} \leftarrow \emptyset.$$

The initially arc-consistent triangle has no two-color solution.

Arc consistency is **local**, not global.

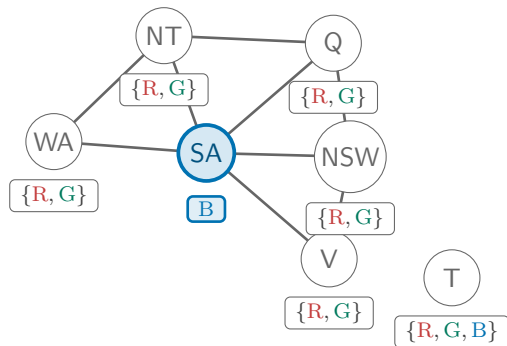
Inference Strength and Cost

Consistency	Locality	Required extension	Implication
Node	1 variable	each value obeys unary constraints	Cheapest preprocessing.
Arc	2 variables	each value has neighbor support	Common; enforced by AC-3.
Path	3 variables	each legal pair extends	Catches the triangle; costlier.
Strong k	up to k	each $k - 1$ assignment extends	Strong pruning; rapidly rising cost.



Strong n -consistency permits assignment without backtracking, but establishing it is exponential in general.

Opening Question Revisited



One assignment, five revisions

$$3^5 = 243 \longrightarrow 2^5 = \mathbf{32}$$

$$\text{reduction} = 1 - \frac{32}{243} \approx \mathbf{87\%}.$$

What powered it?

1. variables expose components,
2. domains expose alternatives,
3. constraints expose unsupported values.

Propagation converts structure into safe search-space reduction.

Constraint-Propagation Takeaways

Represent

$\langle X, D, C \rangle$

Expose variables,
alternatives, and
interactions.

Propagate

delete

Remove values that
have no local
support.

Verify

local $\neq$ global

Arc consistency
need not imply
satisfiability.

Balance

infer vs. search

Stronger reasoning
costs more per
node.

This deck

Model and propagate



Next deck

Backtracking, ordering, and inference

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.